

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1177

Bloxoms Corner 138 kV
August 19, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1177. The Interconnection Customer proposes a 123.1 MW battery energy storage facility interconnecting at the Bloxoms Corner 138 kV substation in the Virginia area. The request was received on March 27, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 389 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$961,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Sewell'S Pt / Yamato 138 kV is overloaded to 113.5% of its 265 MVA emergency rating for the loss of the Shellbank / Kem 138 kV L. The project contributes 32.7% DFAX (40.3 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sewell'S Pt / Yamato 138 kV Line	Rebuild line with higher-capacity conductor	\$2,622,000	37% (\$961,000)
	Total allocated to project		\$961,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.